

Progress Test 1.A — Thermo-S1

1.A · System and state variables

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the molar volume $V_m = 24.5 \text{ L mol}^{-1}$ into $\text{m}^3 \text{ mol}^{-1}$. Give the exact result, without rounding.

[illegible]

Marking

1	
0	1

Exercise 2. Closed and isolated systems — definitions

(5 marks)

Define a closed system and an isolated system. For a gas enclosed in a cylinder, say what the walls must be like to prevent each kind of exchange with the surroundings. Is every closed system isolated?

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Classifying quantities and systems

(4 marks)

Which of these quantities is intensive?

- ☐ the mass m
☐ the volume V
☐ the molar volume $V_m = V/n$
☐ the energy E

A hot-air balloon, with its opening at the bottom, is:

- ☐ closed
 ☐ open
 ☐ isolated
 ☐ closed and isolated

The SI dimension of pressure is:

- ☐ $\text{ML}^{-1} \text{T}^{-2}$
☐ $\text{ML} \text{T}^{-2}$
☐ $\text{ML}^2 \text{T}^{-2}$
☐ $\text{ML}^{-2} \text{T}^{-2}$

 R is an intensive constant. Which relation is homogeneous in the extensive / intensive sense?

- ☐ $\frac{p}{T} = \frac{R}{V}$
☐ $pV = RT$
☐ $\frac{p}{T} = \frac{nR}{V}$
☐ $p = nRT$

Exercise 4. Composition of a solution

(4 marks)

A mass m of sodium chloride (molar mass M) is dissolved in water to obtain a volume V of solution. Compute the amount of substance n of solute, the mass concentration c_m and the molar concentration c . Check that $c_m = cM$.

$m = 5.85 \text{ g}$	$M = 58.5 \text{ g mol}^{-1}$	$V = 0.500 \text{ L}$

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1

— End of the test —